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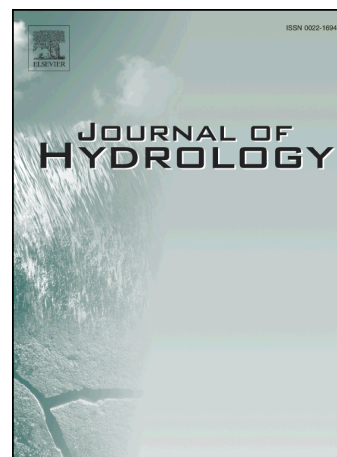
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A participatory system dynamics modeling approach to facilitate collaborative flood risk management: a case study in the Bradano River (Italy)

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Abstract

Participatory methods to support successful policy decisions regarding the complex and dynamic interactions of social, ecological, and physical processes involved in flooding must be simple, easy-to-use, and cost-efficient. Accordingly, a stepwise methodology framework based on causal loop diagrams (CLDs) was developed to address the challenge of context-sensitive initialization of key stakeholders in the collaborative flood risk management process. The methodology consists of five main stages: (i) problem definition, (ii) stakeholder analysis and identification of key groups, (iii) interviews with key stakeholders to construct individual CLDs, (iv) merging of individual CLDs to form a holistic qualitative model representing the entire system, and (v) implementation of an order-oriented reduction process to simplify the final merged CLD, thereby increasing understanding of the most important processes and feedbacks. The proposed approach for flood risk management was tested in a coastal area of southern Italy, a region historically affected by flood events. Given its simplicity, the proposed method emerges as a valuable tool to elicit and map mental models, especially when working with the most inexperienced stakeholders, *i.e.*, farmers, agricultural companies, tourist complexes holders, and infrastructure authorities. Indeed, all stakeholders in the current study were able to understand the process and proposed different flood risk management policies, such as land-use changes, management of damage insurance payouts, floodplain activities, and improvement of public awareness. The proposed methodology overcame multiple barriers in initializing stakeholder engagement, including the technical focus of most flood management agencies, the additional cost and time requirements for stakeholder involvement, as well as institutional structures that impede collaborative management. Moreover, the results point to socio-economic aspects of flood risk management that have not been considered in previous modeling studies.

Keywords

Causal loop diagram; flood risk management; system dynamics; participatory modeling; stakeholder engagement; decision-making.

1. Introduction

Floods are the most frequently occurring and costliest natural hazard throughout the world, killing more than 25,000 humans annually and affecting annually 520 million people and

leaving 3.2 million homeless (Simonovic, 2009). European countries are no exception; between 1990 and 2008 the European Environment Agency estimated that 213 European flood events affected more than 3 million people and caused 1126 fatalities, along with at least US \$52 billion in losses. In the same period, Salvati et al., 2010 reported 2321 flood events in Italy that led to approximately 1407 deaths and 2.8 million injuries. Human activities (*e.g.*, waterway modification, catchment deforestation, and floodplain development) undertaken in natural environments may lead to adverse environmental effects (*e.g.*, the meandering of rivers and blockage of waterways). To address the multiple consequences of flooding caused by human-environment interactions, sustainable stakeholder-proposed management policies must be developed with a holistic understanding of the complex social-economic-environmental context. Misleading or fragmented decisions regarding structural and non-structural interventions can affect flood risk in up- or down-stream communities in ways that are difficult to predict or even demonstrate following a flood event (Thaler et al., 2015). Flood risk is exacerbated by population growth and urban development in vulnerable areas (Nicholls et al., 2008) and therefore, such societal changes must be considered in efforts to improve the sustainability and resilience of risk-based flood management practices (*e.g.*, Hall et al., 2003; Hooijer et al., 2004). Resilience-based approaches require an understanding of the complexity and interconnections of a system, for example how a disruption to one system component can affect various others over time (Linkov and Trump, 2019).

Many flood risk assessment tools of different levels of complexity are available to quantify – albeit with uncertainty – current risk and its evolution (Albano et al., 2017a; De Moel et al., 2012; Merz and Thielen, 2009; Penning-Rowsell et al., 2005). Risk quantification can also be achieved by estimating flood damages and completing uncertainty assessments in the context of flood risk management studies (Jongman et al., 2012; Amadio et al., 2016; Morales-Torres et al., 2016; Albano et al., 2017b; Albano et al., 2018). While methods used

to estimate the loss of life incurred during flooding events have been extensively investigated and reviewed (Escuder-Bueno et al., 2012; Jonkman, 2016), few studies have examined the development of methods for the integration of stakeholder knowledge into the identification of local flood risk management options (Maskrey et al. 2016). Stakeholder participation is important for successful policy decisions (*e.g.*, Saadat et al., 2011; Halbe et al., 2014; Medema et al., 2014; Inam et al., 2017a; Inam et al., 2017b; Inam et al., 2017c). Risk management strategies may fail in the absence of a participatory approach (Nutt, 2002). For example, Nutt (2002) highlights that a premature commitment to a specific solution and a lack of attention to multiple stakeholder requirements can lead to the increased probability of erroneous decision-making. Moreover, flood management policy has a more direct impact on stakeholders than policies in many other environmental fields; floods pose a direct threat to property, economic activity, and human life (Challies, 2016). Accordingly, the engagement of stakeholders in socio-economic-environmental modeling processes is key to the development of a holistic qualitative model, adopting, for example, a system dynamics modeling approach that is capable of considering all socio-economic components and ensuring that all issues and relevant policy opinions have been addressed, as showed in Inam et al., 2015 that have used causal loop diagrams (CLD) as an excellent way of capturing the views and ideas of stakeholders within a model structure. Having analyzed the principal causes of decision-making process failures and methods of avoiding bad practices, Nutt (2002) concluded that roughly half of policy decisions failed to achieve meaningful results because the knowledge and interests of key stakeholders were excluded. These challenges suggest that a flexible approach is needed to address the right problem at the right scale and scope and present the correct system response (Winz et al., 2009). An effective approach must be simple, intuitive, able to support the understanding of complex problems and allow for improvements in the decision-making process.

The three main objectives of the present study are to: (i) propose a simple and detailed approach to identify stakeholders and involve them in the decision-making process; (ii) use qualitative causal loop diagrams (CLDs) to incorporate stakeholder viewpoints and preferred stakeholder policy options through participatory model building; and (iii) explore the application of the proposed approach on the Ionian Sea coast crossed by the Bradano River in the Basilicata region of Italy.

Existing studies have identified the importance of stakeholder input and explored obtaining it in various ways. In addition, system dynamics (SD) models have been used to address other flood-related problems (see section 1.1). However, the current study represents the first time that a methodology based on the use of qualitative models of stakeholders' mental maps has been applied in flood risk management. CLDs were chosen for the current analysis as they provide a flexible qualitative modeling process and allow for the inclusion of both socio-economic and environmental variables, supporting the exploration of the dynamic linkages among variables from both domains (Videira et al., 2009). Beyond capturing stakeholder viewpoints and their preferred management options, CLDs provide a holistic picture of the complete system by representing the interactions between different variables and aiding in the understanding of feedbacks between different system components.

Holistic models developed by merging individual stakeholder-built CLDs are quite complex (Inam et al., 2015) and might limit the stakeholders' role in later modeling stages. Any barrier to stakeholder engagement during the participatory modeling process can lead to bias in the final model (Inam et al., 2015). The approach presented in the current study overcomes these limitations by proposing a methodology that reduces the complexity of the final merged CLD, rendering it more comprehensible for all types of stakeholders, including those with limited modeling and technical skills.

This study offers an important contribution to risk-management decision-making research through demonstrating how stakeholders with limited technical skills, financial resources and time can be involved in flood risk management, ultimately improving knowledge and the decision-making process. Other practical benefits include, but are not limited to, introducing alternative ethical perspectives in decision-making, problem definition, and sustainable management solutions, broadening the range of potential solutions, spreading the responsibilities and risks associated with management actions, and encouraging more balanced decision-making (Parson, 2000; Smith and McDonough, 2001; Sinclair, 2003).

1.1 Background

Many proponents and practitioners of the participatory approach (*e.g.*, Carmona et al., 2013; Voinov et al., 2017) have highlighted the benefits of stakeholder involvement in modeling and decision-making processes. Several recent studies have analyzed different aspects of participatory flood risk management. For example, Thaler et al. (2016) conducted a study in England, exploring the relationship between national and local stakeholders by using a heuristic-analytical framework. Edelenbos et al. (2017) investigated the role and impact of stakeholder initiatives in flood risk management by comparing case studies to find contextual and situational similarities and differences to explain how stakeholder initiatives influence governmental policy. Such approaches, however, have not provided a detailed framework with which to identify and subsequently involve stakeholders; such a framework is fundamental to solving the problems under investigation.

Using a participatory approach based on group meetings, Kotir et al. (2017) assessed water resource management and sustainable agricultural development approaches implemented along the Volta River in West Africa. Using Bayesian networks, Maskrey et al. (2016) similarly implemented a participatory approach to find intervention options for managing

flood risk in Hebden Bridge (UK). In both cases, however, the researchers failed to capture — in a simple, quick, and understandable manner — stakeholders' mutually-uninfluenced opinions related to the specific problem. Instead, they employed mediated modeling, in which key stakeholders worked together in model building. This approach can increase the probability of incurring conflicts of interest by stakeholders that may hinder the success of the application (Halbe et al., 2018). In individual interviews, on the other hand, there is the risk of eliciting uninformed opinions, which must be taken into account in the modeling process. Moreover, the application of Bayesian Networks in the aforementioned study was inherently acyclic, and thus, unable to handle the system's feedback structure (Molina et al., 2010).

Local stakeholder input in environment and development related decisions is supported at international and European levels (*e.g.*, 2007/60/EC Article 10). In particular, the importance of stakeholder involvement in flood risk management has been emphasized in policy and practice in the EU, US, and elsewhere (Wood et al., 2012; Tingsanchali, 2012; Begg et al., 2018). Tingsanchali (2012) explored the effects of integrated urban flood risk management in Thailand. The researcher suggested that the involvement of stakeholder knowledge from different perspectives enables a coherent understanding of flood risks, therefore improving the outcome of decision-making processes. The paper highlighted that the involvement of local stakeholders allowed for the identification and implementation of more effective and sustainable flood management measures that are supported by the majority of stakeholders. Begg et al. (2018) discussed two practical examples in Europe (*i.e.*, the UK and Germany) of local stakeholder participation in flood defense-related decisions. Wood et al. (2012) illustrated the development of an integrated approach to flood preparedness planning that harmonizes stakeholder and agency efforts in the US.

These studies demonstrate the efforts of the research community and flood risk authorities to improve current levels of stakeholder participation in flood risk management, suggesting a trend towards greater openness and inclusion of stakeholders. Despite an emphasis on local stakeholder participation, the actual stakeholders participation and influence is limited in practice, either because participation often takes place at the end of the planning process, or because the participation processes are mainly based on a preexisting decision-making structure (mainly based on economic criteria). Furthermore, stakeholder involvement is often limited due to high time and resource costs. Low potential for stakeholders to significantly influence their role, the outcome, or the starting-point of the decision-making process may lead to frustration and conflict, as well as increased inequality.

The development of a model framework for a cost-efficient stakeholder involvement in flood risk management policies will address previous approaches' lack of institutional support, clear and effective guidelines, communication, information sharing, and resources, especially regarding large stakeholder participation processes (Thaler and Priest, 2014). Research shows that stakeholder engagement at a late stage adversely affects their willingness to be involved in the participatory process, as they often believe the decision-making process has already been completed (Reed, 2008). Based on CLDs (Videira et al., 2009), the present modeling approach is centered around addressing issues that can adversely affect the decision-making process, including a lack of clarity in the objectives and transparency in the engagement process, biased policies that lead to hasty decisions, poor relationships between stakeholders, and an unequal degree of power or level of stakeholder influence. Currently, a few standard approaches exist to involve stakeholders in participatory modeling (Wood, 2012; Begg, 2018). Normally, stakeholders are brought into the process through group meetings or interviews in predefined forms (Mayer et al., 2017). Such engagement may result in biased models due to the presence of group pressure (Maskrey et al., 2017). Accordingly, for any

participatory process to be successful, these factors should be well defined from the outset (Cornell, 2006; Reed, 2008).

In practice, the participatory modeling process is facilitated through the use of different types of analytical and system tools. The most common and frequently used tools include SD, Bayesian belief networks, fuzzy cognitive mapping, and agent-based modeling (Voinov, 2010).

The main advantage of SD modeling approaches is that they offer a powerful method for explicitly incorporating both social and physical processes into dynamic models, thereby capturing important feedbacks and connections between two types of system variables (socio-economic and environmental) that may lead to counterintuitive system behavior (Sterman, 2000; Halbe and Adamowski, 2011). The proposed study focuses on SD approaches because they provide room for local and expert knowledge in the analysis of environmental issues.

SD has a long history as a modeling paradigm, with its origins in the work of Forrester (1989), who developed the subject to provide an understanding of strategic problems in complex dynamic systems. By providing insight into feedback processes, SD models provide system users with a better understanding of dynamic system behaviors. Areas of SD application have always been broad, with an emphasis on socio-economic applications. SD models have been successfully used for water resources planning and management, policy analysis, sustainable development, and natural resource management (Ahmad and Simonovic, 2011; Chen and Wei, 2014; Hassanzadeh et al. 2012; Kotir et al. 2017; Solana-Gutiérrez et al. 2017; Tehrani et al. 2011; Zare et al. 2019).

In particular, participatory SD modeling has become increasingly popular and effective as it integrates the benefits of SD modeling and stakeholder involvement. Participatory SD modeling variable relationships are initially constructed using qualitative information-gathering techniques, such as surveys, questionnaires, group model-building exercises, or

story-telling. The results of a participatory model-building effort include the improved understanding of (i) cross-variable interactions, (ii) study system complexity, (iii) stakeholder dynamics and socio-cultural norms, (iv) management limitations, and (v) potential authoritative hierarchies. The benefits of performing a participatory SD modeling assessment can be physical, environmental, social, or economic. For example, Pagano et al. (2019) proposed a participatory SD modeling approach for flood risk reduction with the additional goal of producing stakeholder-friendly livelihood benefits, such as nature conservation and improved community well-being. Participatory SD modeling has also been used in the industrialized pastoral farming (Puig et al., 2011), integrated coastal zone management (Mongruel et al., 2011), and interdisciplinary water systems management (Sunsnik et al., 2012).

Although participatory SD modeling is a useful and effective approach for incorporating stakeholder knowledge into the model-building process, it is still a relatively new modeling technique, and thus, has some limitations. Many participatory SD modeling models are characterized by varying degrees of uncertainty involving model assumptions, lack of reliable and consistent household panel data, lack of spatial information, or the absence of a robust evaluation for selected coping and improvement strategies. Furthermore, by making the interconnectedness of a modeled environment more explicit, there is a risk of igniting or exacerbating resource or land-use conflicts among beneficiaries (Voinov and Bousquet, 2010). Despite these apparent limitations, participatory SD modeling is a valuable tool that allows for improved comprehension of a study system, increased involvement of local ideals and knowledge, and enhanced decision-making.

Different SD techniques, including CLDs, influence diagrams, fuzzy cognitive maps, and system archetypes, can be used to formulate and represent a dynamic hypothesis (Elsawah et al., 2017; Mirchi et al., 2012). As simple and commonly-used diagramming tools, CLDs are

generally employed at the beginning of socio-economic-environmental modeling exercises to develop a preliminary dynamic hypothesis (Inam et al., 2015). System archetypes are high-level conceptualizations that provide insights about common patterns of behavior in systems (Wolstenholme, 2003). These insights can be transferred across various problem situations and domains (Gohari et al., 2013). In contrast, influence diagrams represent a decision situation achieved through the generalization of Bayesian networks, providing a more explicit and rigorous graphical representation of the causal structure than CLDs. As an alternative, fuzzy cognitive maps, a semi-quantitative diagramming tool that represents a belief system about the problem's causal structure, allow one to portray the relative strength of inter-variable links (Kok, 2009). However, influence diagrams have a lot of conventions and rules, making them time-consuming to master, while the output of fuzzy cognitive mapping can be difficult to explain to non-modelers (Van Delden et al., 2008). Furthermore, influence diagrams are difficult to examine from different perspectives, increasing the risk of force-fitting the problem situation system to an archetype (Gohari et al., 2013). According to Elsayah et al. (2017), this risk is exacerbated if the modeler is inexperienced (Corben, 1994). Comparatively, CLDs provide an aggregate view of the problem structure, maintaining focus on feedback loops instead of unnecessary details. In addition, CLDs are relatively intuitive and easy to explain to non-modelers (Sedlacko et al., 2014).

2. Case Study

Crossing southern Italy's Basilicata region from the northwest to the southeast and ending in the Gulf of Taranto, in the Ionian Sea, the 170 km Bradano River extends over a total surface area of 2960 km² at an average elevation (AMSL) of 387 m. The western and southwestern portions of its basin are mountainous (with elevations ranging between 700 and 1250 m), while the southern downstream portion, addressed in the present study, does not exceed 500

m in elevation (*i.e.*, the Metaponto Plain). The river's hydrographic network is ramified, with its main tributaries entering from the northeast side of the main river course. The Bradano River basin has three main sections: (i) a top section, upstream from the San Giuliano dam; (ii) a middle section, downstream from the dam but upstream from Route SS 106; and (iii) a bottom section, which is the most flat area of the river flowing into the Ionian sea (see the left portion of Fig. 1).

During recent flood events, the greatest damage to agribusinesses, farms, and tourist complexes were recorded downstream from the San Giuliano dam. During the latest flood event, the main road (SS 106) and railway were closed to traffic for several days. Three critical issues have been identified by simulating a flood event with a 30-year return period (see Fig. 1):

1. In the fluvial reaches upstream from Route SS106, the northeast hydraulic bank is not sufficient to retain a 30-year flood, which can then extend for about 2 km to the northeast, reaching ground areas located downstream.
2. In the fluvial reaches between Route SS 106 and the railway embankment, water, in a 30-year flood event, at depths exceeding 1 m would be mainly concentrated in a narrow strip of land upstream from the railway, while the rest of the flooded area would be under less than 1 m of water.
3. Downstream from the railway embankment, banks on both sides of the riverbed are insufficient to retain a 30-year flood event; moreover, underpasses favor the outflow of floodwaters from upstream to downstream of the railway.

This portion of the Ionian coast, known as the Metaponto plain, is subject to critical hydraulic conditions due to a network of drainage channels designed to collect and convey waters into to other areas. These channels, managed by the region's land reclamation consortia, provide the only artificial drainage for surface water and/or outcrops in the area. Finally, in the

downstream section near the sea, there are protected areas associated with the archaeological remains of early settlements dating back to the Archaic period.

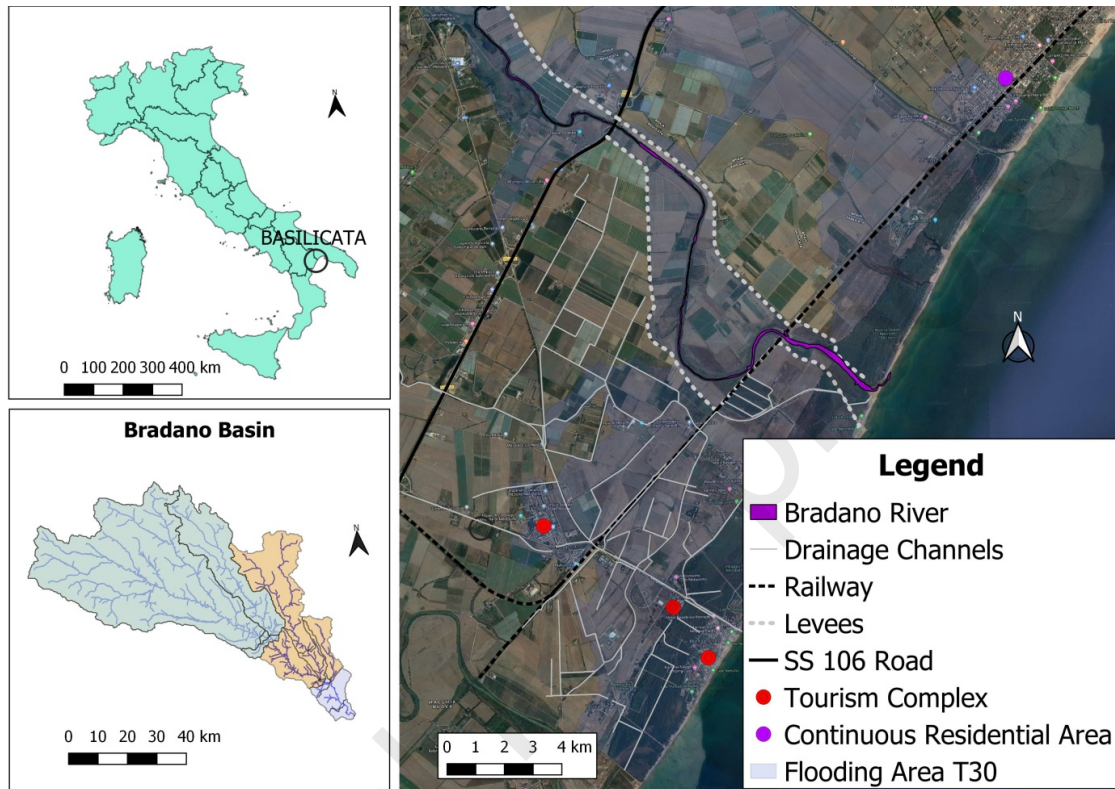


Fig. 1 – Study area and results of the simulation for a flood event with a 30-year return period.

Historically, the Bradano basin has been affected by flooding events (*e.g.*, recently in October 2010, March 2011, and December 2013) that cover large areas and damage infrastructure, productive activities (agriculture and tourism), and archaeological heritage sites (Scarpino et al., 2018).

In the 2011 flood, all rivers in the Basilicata region (the Agri, Sinni, Basento, Cavone, and Bradano) flooded, requiring the evacuation of 200 families from the city of Matera and leading to the destruction of hundreds of hectares of crops, the closure of farms and zoo-

technical enterprises and the death of hundreds of livestock. The southeastern portion of the region was the most severely affected.

The 2013 flood in Matera province occurred following a pair of meteorological events which occurred two days apart (October 5 and 7, 2013) and resulted in total precipitation of 186 mm (Table 1). This flood, which affected the entire coast, caused serious and widespread damage, and killed four people. The Bradano River and its tributaries showed extensive flooding, with floodwaters overtopping bridges across the riverbed. The railway line was interrupted, and extensive damage to agriculture and housing infrastructure was recorded. The archaeological park near the river's outlet was submerged by water that had overflowed from adjacent drainage channels.

The hydro-meteorological event that occurred during December 2013 (between November 30, 2013, and December 3, 2013) mostly affected the southeastern part of the Basilicata Region. This event was particularly intense due to the rainfall's high intensity; the gauge stations in the area registered cumulative rainfall over 150 mm (Manfreda et al., 2015). The rainfall gauge station in Metaponto town exceeded 200 mm. The flooding of the Bradano and Basento rivers and the overflow of the reclamation channels caused the interruption of several provincial roads and the highway SS 106. Moreover, the flooding impacted the inhabited centers of Policoro and Mataponto, as well as the archaeological park and agriculture activities (Manfreda et al., 2015). Figure 2 shows the flood hydrograph of the December 2013 flood. At about $800 \text{ m}^3 \text{ s}^{-1}$, the recorded peak discharge was less than the discharge calculated ($1736.53 \text{ m}^3 \text{ s}^{-1}$) for a 30-year return period. The discharge expected for a 30-year flood event causes an extension of the flood-prone area, as shown in Fig. 1, which demonstrates that the current hydraulic structures along the watercourse are insufficient to contain water flow with a return period of fewer than 30 years. Table 1 shows the cumulative precipitation values recorded during the three recent flood events.

Stations	Cumulative precipitation (mm)		
	Flood events		
	1 st Mar 2011	6 th and 7 th Oct 2013	1 st Dec 2013
Torreaccio	104	186	134
San Giuliano	160	102	136
Montescaglioso	115	137	144

Table 1: Cumulative precipitation recorded during the 2011 and 2013 flood events (Manfreda et al., 2015)

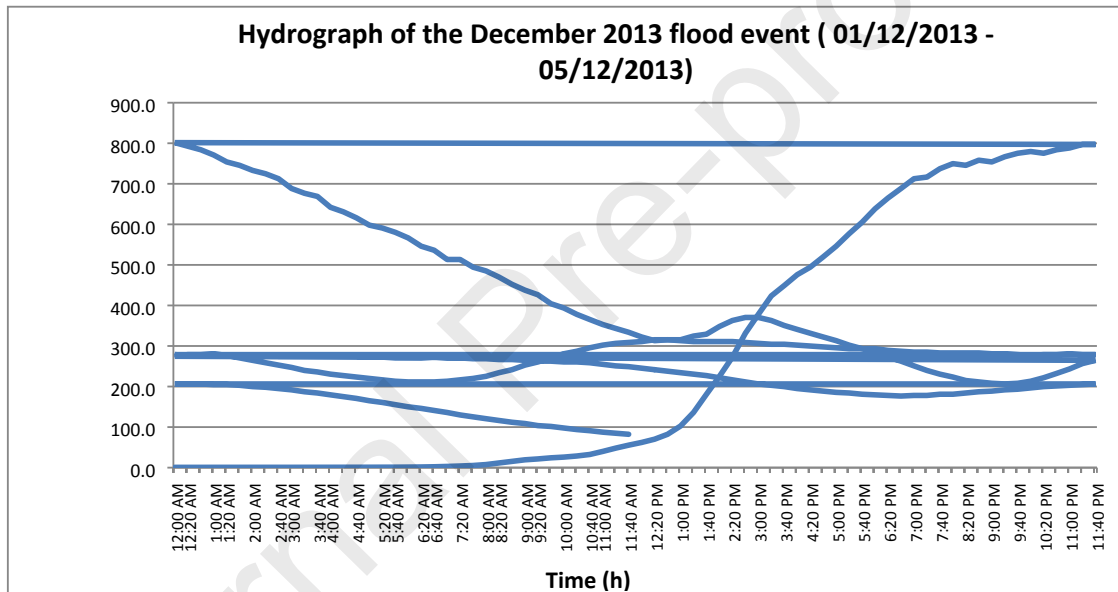


Fig. 2 – Hydrograph of the December 2013 flood event

These recent flood events, along with pressure from local media and farmers regarding the potential role of the dam in flood consequences, have led the Basilicata Watershed Authority to adopt a dam management plan. Specifically, the plan aims to increase the lamination effect of the dam, without structural changes, thereby reserving a specific volume of the dam's reservoir to accommodate the water transiting during a potential flood event, allowing for the water's gradual release, decreasing the potential peak discharge and, hence, the potential damages (Autorità di Bacino della Basilicata, 2015). In particular, after the Watershed

Authority commissioned single and two-dimensional hydraulic modeling (Sole et al., 2013) for the “Assessment of the effects of hydraulic risk mitigation measures in the final stretch of the Bradano and Basento rivers,” 20 Mm³ of the dam volume was dedicated to mitigation of potential flood events (Autorità di Bacino della Basilicata, 2015). Moreover, a structural intervention related to the enlargement of the bridge culvert of the main railways, Taranto-Reggio Calabria, is already underway based only on the technical studies lead by the Basilicata Watershed Authority (Autorità di Bacino della Basilicata, 2015).

Although many farmers, tourism complexes holders, and citizen living in the area at risk have made some pressure thought decision-makers thanks to local media in order to solve the problem of flooding in the region, their involvement in the decision-making processes has been limited to few public meeting based on preexisting decisions.. Therefore, the current solution may neglect the socio-economic aspects of flood risk reduction specific to the study area. This suggests that important details and potentially useful information for reducing the flood risk in the area may have been overlooked during the decision-making process. The identification of this oversight is important as it indicates an insertion point for steps to improve regional flood management in the future.

3. Methodology

Following the principles of system thinking and SD (Forrester, 1961; Sterman, 2000), CLDs are employed in the present study to construct conceptual models to support flood risk management decision-making. CLDs were chosen for their ability to show cause-effect relationships between the variables that characterize a dynamic system through a simple graphical structure. If simplified, CLDs can easily be understood by non-technical users due to their graphical nature, making them an ideal modeling tool in a participatory setting (Kotir et al., 2017).

Depending on the composition of the stakeholder groups and the specific case under analysis, various stakeholder engagement tactics can be adopted (Voinov et al. 2010). In the present study, meetings were carried out with single individuals to avoid conflicts between opposing stakeholder groups and facilitate an open and neutral discussion. Stakeholders tend to have less interest in participating in group meetings than in individual interviews (Burgin et al., 2013; Videira et al., 2009), since it is often more difficult for individuals to express their views openly in a group setting due to the presence of ‘opposing’ groups or superiors in their organization. For this reason, group meetings were omitted in favor of individual interviews. Problem definition is often considered the most important step in model development because the clear articulation of the problem is essential for defining the purpose of the model, its boundaries, and time domain. At the same time, problem definition affects the selection of stakeholders in the modeling process. In the current study, the problem definition step was carried out by the facilitator through literature review, discussion with local stakeholders, and the study of historical maps. After problem definition, the selected problem was considered in the context of stakeholder needs and its ability to maintain stakeholder interest throughout the modeling exercise. A widely tested framework developed by Elias et al. (2002) and improved by Inam et al. (2015) was used, whereby, once stakeholder analysis was complete, the development of individual stakeholder CLDs was facilitated in a short period and with minimal financial resources. Finally, the individual CLDs were compared and merged into a composite diagram. The resulting diagram was discussed in a group stakeholder meeting to create the final group CLD. Using the methodology of Bureš (2017), the composite CLD was reduced and simplified to make the final CLD understandable to all stakeholders.

The overall methodological framework (Fig. 3) used in the present study consists of four main stages: *Stage I*, problem definition; *Stage II*, stakeholder analysis; *Stage III*, interviews with stakeholders and construction of the individual CLD models followed by model

digitization using the Vensim software (<https://vensim.com/>); *Stage IV*, construction of a complex group CLD; and *Stage V*, simplification of the merged CLD model. The following sections provide a detailed description of each step.

3.1 Problem definition

Clear articulation of the problem, its boundaries, and the timeframe within which it is to be addressed is critical to the success of the modeling process. The problem must be defined before proceeding with stakeholder analysis and CLD modeling. A well-prepared and clear definition of the problem aids in stakeholder selection and the definition of clear objectives, making it possible to determine the appropriate level and methods of engagement (Reed, 2008).

The five main phases of problem definition include: (i) selection of the problem theme and key variables; (ii) selection of the time horizon; (iii) definition of model limits and boundaries; (iv) development of reference models (for example, a graphic representation of change over time) that represent the dynamic behavior of the problem; and (v) identification of relevant stakeholder groups.

The definition of the problem is an iterative process; after stakeholder analysis, it can be modified, and even completely changed if stakeholders consider it necessary.

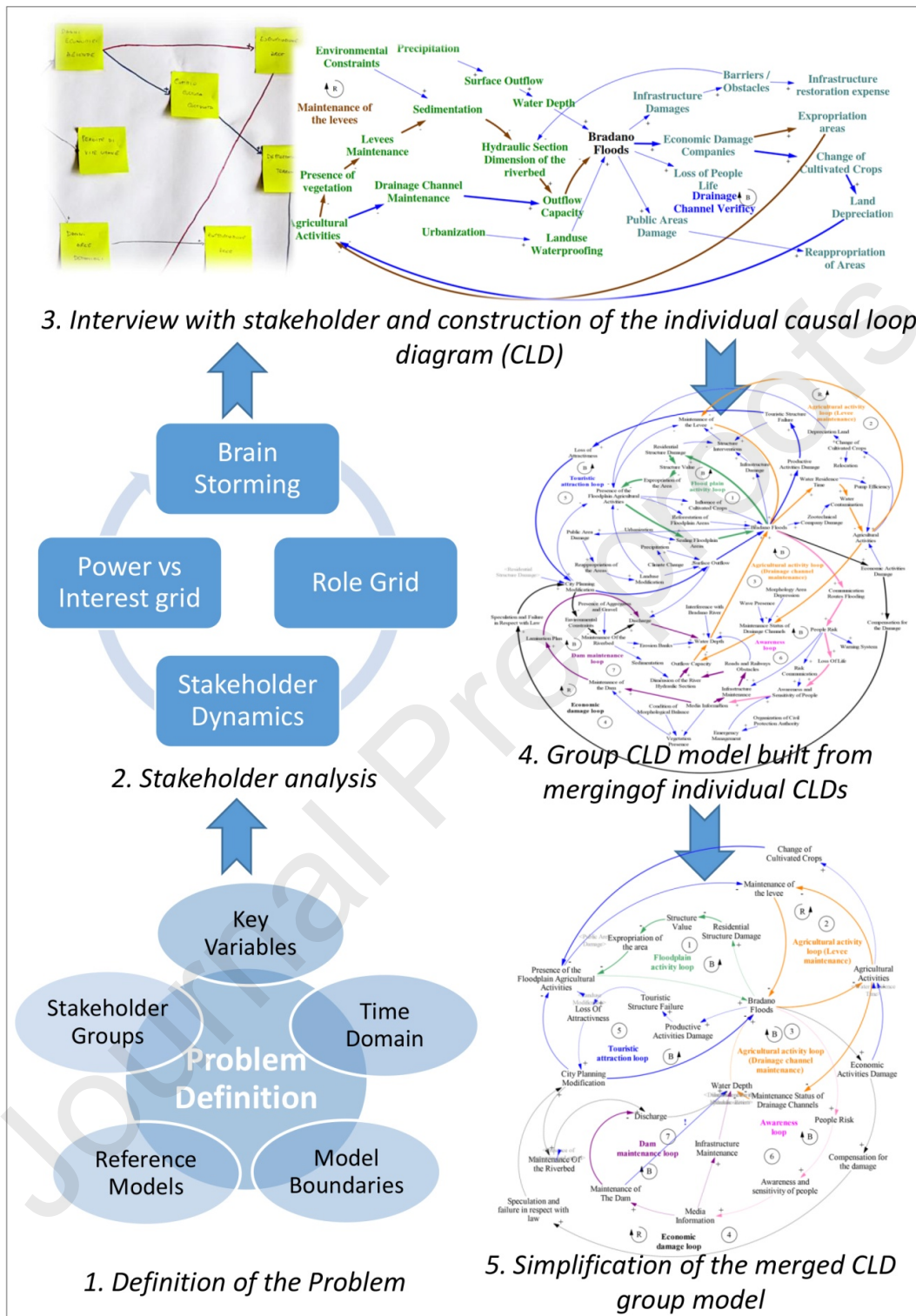


Fig. 3 – Schematic representation of the proposed methodological framework

3.2 Stakeholder analysis

Once the problem has been defined it is necessary to identify the stakeholders who should be involved in the modeling process. Several methods of carrying out stakeholder analysis exist (Brugha and Varvasovzky, 2000; Mitchell et al., 1997; Elias et al., 2002; Reed et al., 2009) and the approach varies according to the type of problem. For example, Lindenberg and Crosby (1981) suggested making an inventory of those who could have a role in decision-making, gauging their importance through their level of influence and their interest in a particular outcome, mapping the relationships between these actors, and understanding their potential for developing alliances. Stakeholder analysis is also used to understand the diverse range of potentially conflicting stakeholder interests. Therefore, the process of stakeholder analysis can exacerbate and generate conflict. In some cases, stakeholders may have hidden agendas or covert interests may also skew the analysis, expressed concerns over research objectivity since those undertaking the analysis often do so from a particular perspective or with particular outcomes in mind. The current study uses the framework proposed by Inam et al. (2015) to render the resulting structure more complete, inclusive, and versatile than other approaches. The process occurs in four main phases: (i) listing all stakeholders, including marginal stakeholders; (ii) categorizing them according to their roles; (iii) prioritizing them according to their attributes; and (iv) final selection based on the developed power and interest grid.

A preliminary list of stakeholders can be effectively developed using an approach that identifies relevant stakeholders either through a literature survey or based on the individual's local knowledge. In the current study, interested stakeholders were then be asked to identify other relevant stakeholders. This process (also termed brainstorming) may involve seemingly marginal stakeholders (Brugha and Varvasovzky, 2000). To find gaps in the first brainstorming step and to look for relevant parties who may have been omitted, the list is

further categorized according to each stakeholder's role, as defined by the European Commission (2003), as decision-makers, users, implementers, or experts.

Following the methodology of Mitchell et al. (1997), the interested stakeholder is further classified according to three relational attributes, *i.e.*, power, legitimacy, and urgency. Entities with no power, legitimacy, or urgency concerning the problem are not stakeholders and are perceived as having no salience. Finally, the stakeholder analysis is finalized by creating a power vs. interest grid (Crosby and Bryson, 2005) that consists of a map showing the level of interest of each key stakeholder regarding the problem, in relation to their power. Notably, during the process, stakeholder attributes may evolve in light of modifications to the state of the system or strategic issues. In this way, the final selection of stakeholders involves representatives in all roles (*i.e.*, decision-makers, users, implementers, and experts), who are related to at least one of the attributes of power, legitimacy, urgency, or interest.

3.3 Stakeholder interviews for the construction of individual CLDs

After stakeholder analysis is complete, potential stakeholders are contacted for individual interviews that will involve the construction of individual CLDs that express their views regarding the problem to be addressed.

The CLD is a modeling tool developed to represent the complex and dynamic nature of system behavior. It responds to the requirements of the SD approach by treating organizations as systems made up of interacting parts, instead of examining isolated events and their causes. CLDs are excellent for quickly capturing stakeholder hypotheses regarding the causes of SD, eliciting and capturing the mental models of individuals or teams, and communicating the feedbacks that a stakeholder believes to be contributing to a problem. CLDs have been successfully used in water resource planning, policy analysis, sustainable development, and natural resource management (Prodanovic and Simonovic, 2007; Sendzimir et al., 2007).

CLDs also feature feedback loops that are important for understanding SD. A closed cycle is either defined as a reinforcing or balancing feedback loop. A positive feedback loop generates reinforcing behavior that contributes to exponential growth in a system, while a negative feedback loop generates balancing behavior that acts as an equilibrator in a system. The process of CLD development is based on a methodology that provides a useful start-up framework for the group model building approach and addresses important questions about the principal setup of a group model building exercise (Vennix, 1996). The steps of the framework are described below:

1. A stakeholder (*e.g.*, a local farmer) is presented with the purpose of the mental model exercise, and through a simple example, instructed in the method of drawing a CLD. Colored sticky notes and large sheets of plain paper are provided to the stakeholder to allow them to create their mental model.
2. The stakeholder inserts the problem variable at the center of the sheet.
3. The stakeholder is then asked to add what they consider to be the causes of the problem (first the direct or first-order causes, followed by the indirect or second-order causes, and last, any higher-order causes) on the left side of the problem variable, using sticky notes. The causes are joined through causal links, and the polarity and directionality of arrows are added.
4. Next, the direct and indirect consequences of the problem (or the first-, second-, and higher-order consequences) are added to the right of the problem variable. Likewise, causal links with polarities are defined among the consequences.
5. Finally, the causes and consequences are joined through feedback loops.

The following types of questions are asked during the interview to help each stakeholder develop their CLD:

1. How has the problem developed over time?

2. What are the main direct and indirect causes of the problem's development, including link polarities?
3. What are the consequences of the problem?
4. What are the main feedback processes?
5. What kinds of short-term policies do you think could be adopted to solve this problem?
6. What kinds of long-term policies could be adopted to solve this problem?
7. What are the main obstacles to the success of these policies?

Depending on stakeholder availability, one of three interview options is adopted. When time is limited (*e.g.*, 15 minutes), the interview is recorded and subsequently translated into a CLD by the facilitator. The CLD realized by the facilitator is sent to the stakeholder so they may verify whether the diagram represents their point of view.

When the interviewee's availability is slightly longer (*e.g.*, 30 minutes), a preliminary diagram prepared by the facilitator is presented to the interviewer and they are asked to correct the diagram and add missing links. If a stakeholder has sufficient time (*e.g.*, greater than one hour), the modeling exercise is started from scratch and the stakeholder carries out each step.

Through the construction of the CLDs, stakeholders express their views regarding the problem, its causes, its consequences, the feedback loops involved, and strategies and policies that could be applied to address the problem. All policy options are included in the diagram as policy variables. Important loops are numbered and labeled based on their reinforcing or balancing behavior. Furthermore, for quick reference, the loops are named according to the process they represent. After the completion of the stakeholder interview phase, the facilitator translates all CLD models into digital computer models using the SD modeling software, Vensim DSS (<https://vensim.com/>).

3.4 Construction of an overall group CLD

After carrying out all stakeholder interviews and developing the individual CLDs, the facilitator(s) construct a preliminary diagram by analyzing, comparing, and merging all individual CLDs. The composite diagram is then discussed in a stakeholder group meeting to create a final group CLD, shared by all the stakeholders.

The merged model aims to represent the different points of view and mental models/maps of all stakeholders with respect to the problem, its causes, its consequences, the involved feedback loops, and possible strategies and policies that could be employed to address the problem. This makes it possible to highlight the perspectives of different stakeholders and develop a representative mental map of the system, considering both complementary and controversial elements. The resulting combined CLD model includes the divergent perspectives of all key stakeholders (those that were interested in participating) and can be used to describe the qualitative behavior of the system through different reinforcing and balancing loops. This type of qualitative analysis cannot be used to infer quantitative behavior, but serves other purposes, including the detection of the critical problem system, the identification of knowledge gaps for further research, the enhancement of stakeholder understanding, the detection of conflicts of interest, and the facilitation of decisions related to long-term policies. Merging the individual models of several stakeholders can enhance the details of the model so that sub-models on various aspects of the problem can be identified. The proposed study uses the approach of Inam et al. (2015) for generating a comprehensive model that includes the diverging perspectives of stakeholders addressing conflict and controversies that arise during the merging process of individual CLDs.

It is essential, however, to avoid a complex mental model, as complexity makes the model less intuitive and reduces stakeholders' ability to engage and identify the assumptions underpinning the model's causal links. Complexity is an issue in the SD realm, as it restricts

the ability of stakeholders with limited modeling skills to understand complex holistic systems. The current study seeks to develop a flood risk management process that is accessible to a wide variety of stakeholders through the establishment of a tool to simplify the representation of complex systems and make them understandable. This was achieved through the CLD simplification approach suggested by Bureš (2017), explained in the following section.

3.5 Simplification of the overall group CLD

As mentioned above, the combination of the individual CLDs can produce an unwieldy final group CLD model (Sedlacko et al., 2014), leading to incorrect inferences (Mirchi et al., 2012). Moreover, if the analyzed environmental problem involves different administrative regions in addition to different types of stakeholders, further conflicts may arise due to stakeholders with differing viewpoints proposing completely different mitigation actions for the various regions involved.

According to Bureš (2017), by building a simplified CLD model, the complexity of the final CLD is reduced and, as seen in the present work, these limitations can be overcome. A small model is often sufficient to explain problem dynamics and build intuition regarding appropriate responses (Ghaffarzadegan et al., 2011). Therefore, it is necessary to improve the readability of visual system representations. The methodology presented here is based on the initial construction of the CLD using duplicate variables as the input to other variables (Bures, 2017). In fact, single crossing links, which create intersections that add considerably to the complexity of the diagram, have been avoided in order to improve overall diagram readability.

The method is built around three main activities, endogenization, encapsulation, and order-oriented reduction:

- i. The endogenization step involves an initial simplification of the model through the deletion of exogenous system variables to make the diagram fully endogenous.
- ii. The encapsulation step entails the selection and elimination of all variables that have a single input and a single output. Subsequently, based on the principle used to define the feedback loop polarity (Sterman, 2000), a new polarity for the connection between variables is established. These steps are repeated until the model no longer includes individual input and output variables.
- iii. The order-oriented reduction step consists of marking and deleting all single-input double-output variables, along with double-input single-output variables to increase the readability of the diagram.

These simplification actions can be carried out individually or consecutively. In general, endogenization is preferable to encapsulation, and encapsulation is preferable to order-oriented reduction.

Once the reduced CLD is obtained, the facilitator visits stakeholders to verify that the merged diagram still represents their point of view, despite the simplification and elimination of variables.

4. Results

The Bradano River case study presented here investigates the level of stakeholder engagement in evaluating the causes, consequences, and policies for flood management through a participatory modeling approach. From the onset of the study's preliminary interview phase (individual CLD development with each stakeholder), stakeholders confirmed that flooding of the Bradano was a critical problem for the region (see Section 4.3). Without requiring that the interested groups have any technical experience, the CLD methodology proved to be effective, in terms of both cost and time. After an introductory 20-

to 30-minute session, nearly all stakeholders were able to apply the method and build a CLD from scratch. The practical application of the methodology is discussed in the following section.

4.1 Stage 1: Problem definition

Due to the massive damages caused by recent flood events, flood risk in the area under investigation is receiving close attention from stakeholders and the media. The particularly intense meteorological phenomena that affected Basilicata between the period 2010-2013 contributed to hydrogeological issues such as the flooding of rivers and streams, as well as landslides. This resulted in serious damage to infrastructure and public and private buildings. In such cases, timely and informed decision-making becomes increasingly important. An innovative modeling tool, able to simulate socio-economic and environmental interactions through feedbacks to propose sustainable management solutions can be applied to enhance decision-making abilities. The qualitative modeling approach proposed in the current study addresses the communication gap between the authorities and stakeholders and provides a management solution for flooding along the Bradano River, as defined through the engagement of interested stakeholders.

4.2 Stage 2: Stakeholder analysis

Stakeholder analysis is the process of categorizing and filtering stakeholders based on their roles, attributes, and interests. In the first step, a preliminary list of stakeholders was created by the facilitator through a brainstorming process. The preliminary list comprised all stakeholders, including those considered to be marginal.

A literature review and expert opinion showed that the Basilicata Watershed Authority and San Giuliano Dam Management Authority were important stakeholders in the Bradano flood

analysis. The dam plays a crucial role in the management of water resources, and some stakeholders believe it was the main cause of flooding in 2011. The Watershed Authority establishes the perimeter of flood-prone areas for all the rivers in the Basilicata region and oversees hydraulic risk management, documentation, and projects. The Organization for Damage Assistance for residents was contacted, as it has been involved in helping residents affected by past events. The road authority was also considered since such infrastructure reduces the outflow capacity of the riverbed during flood events. The Early-Warning Authority was considered due to its role in post-event flooding emergency management, and since it collects data regarding precipitation. Farmers who suffered flood damages in past years were also important stakeholders. Other stakeholders were the regional agricultural department, local authorities (e.g. municipality), and tourism authorities.

In the next step, stakeholders were categorized as experts, decision-makers, implementers, or users, using the framework developed by the European Commission (2003). Such categorization helps in the assessment of missing links in the initial stakeholder identification. The stakeholder categories and their roles are summarized in Fig. 4.

After stakeholder identification and categorization, stakeholders were contacted. The preliminary list was shared, and identified stakeholders were asked for suggestions about potential missing stakeholders. Most contacted stakeholders were satisfied with the list but suggested adding the Water Cycle Authority, as they thought that it might help provide further information about public areas close to the river. Stakeholders were then categorized based on their attributes. This step helps to track changes in stakeholder dynamics over time; a stakeholder's changing role can be measured by the loss or gain of one or more attributes. For this purpose, the complete methodology developed by Mitchell et al. (1997), which classifies stakeholders based on three relational attributes (power, legitimacy, and urgency), was used (Fig. 5).

Definitive stakeholders were classified as those who possessed all three attributes. These stakeholders were considered to be the most important stakeholders to include in the qualitative modeling exercise. In the current study, the definitive stakeholders were the Early-Warning Organization, the Regional Agricultural Department, the Basilicata Watershed Authority, the San Giuliano Dam Management Authority, and the infrastructure management authority.

DECISION MAKERS

- Basilicata Region Authority
- Infrastructure Management Authority
- Local Authority
- Regional Agricultural Department

EXPERTS

- Weather Alert Organization
- Watershed Of Basilicata Authority
- Government Institution For Waste And Water Resources
- Dam Management Authority

USERS

- Damage Assistance Organization For Residents
- Tourism Complex
- Farmers
- Railway Authority
- Roads Authority
- Regional Agency For Environment Protection
- Irrigation Authority

IMPLEMENTERS

- Roads Authority
- Tourism Authority
- Railway Authority
- Weather Alert Organization
- Dam Management Authority

Fig. 4 – Preliminary list of stakeholders and their respective roles

In addition, the power vs. interest grid approach, developed by Crosby and Bryson (2005), was completed to prioritize stakeholders. Fig. 6 reveals that the Basilicata Watershed Authority, the regional agricultural department, and the San Giuliano Dam Management Authority have high power and high interest levels. Hence, they were considered the most important stakeholders with regards to flood management in the study area.

Finally, based on stakeholder analysis and stakeholder recommendations, the Basilicata Watershed Authority, the San Giuliano Dam Management Authority, local farmers, the

organization in charge of damage assistance for residents, and the Infrastructure Management Authority, were included in the individual CLD modeling process.

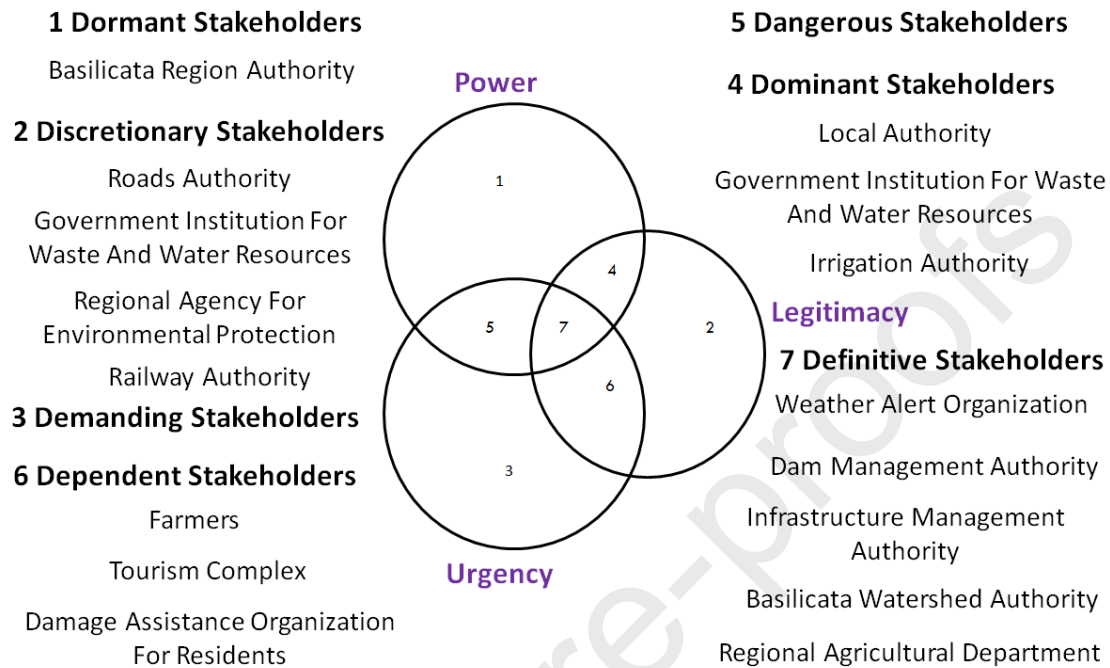


Fig. 5 – Stakeholder typology with attributes

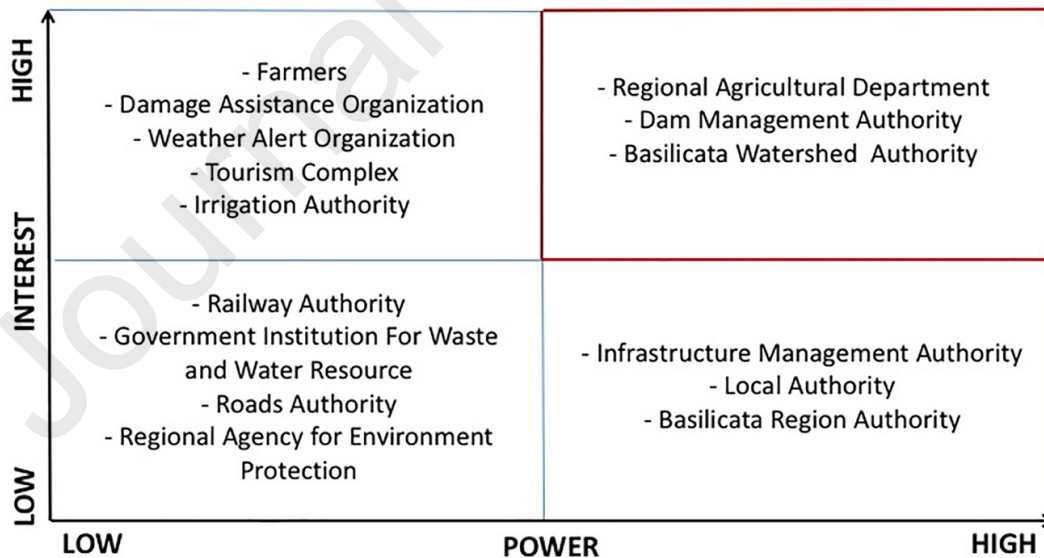


Fig. 6 – Stakeholder's Power vs. Interest Grid

4.3. Stage 3: Stakeholder interviews and individual CLD preparation

During the processing for developing individual CLDs, the authors of this paper facilitated the interview process, which involved meeting with individual stakeholders over a period of eight weeks. In general, the stakeholders showed confidence in the modeling exercise and were satisfied with the representation of their mental maps through the CLDs.

Individual CLD modeling begins with the selection of a problem variable. When stakeholders were initially asked to point out the main problem in the study area, based on the damage inflicted by past events the majority pointed to the flooding of the Bradano River. The next step involves adding direct and indirect causes for the flooding in the study area. Different stakeholders expressed different perspectives, but, generally, they identified water depth exceeding safe limits as the main contributor to flooding. Some stakeholders believed that inadequate maintenance of the riverbed and levees to be one of the primary causes of the floods. In their opinion, these structures created obstacles, effectively reducing the river channel's hydraulic section so that the outflow capacity of the river was insufficient to contain the rising water. Some stakeholders pointed to the management of the San Giuliano dam, emphasizing the need to improve dam management and highlighting the fundamental role the dam has had in past flood events. However, this view contrasted with that of stakeholders who viewed the dam management as satisfactory in terms of irrigation and flood water management. Other stakeholders believed that the railway and Route SS 106, which cross the river (see Fig. 1), contributed to a reduction of the river's hydraulic section by presenting obstacles to river outflow, thus causing water depth to exceed safe limits. Another cause highlighted by stakeholders was the maintenance of the drainage channels that interfere with the Basento river, which is located near the study area.

The next stage in the CLD-building process was the stakeholder-led identification of significant direct and indirect consequences of the Bradano River floods. Major consequences

included various economic, environmental, social and agricultural damages, for example, loss of life, reduction of production, damage to economic activities, farms and zoo-technical enterprises, infrastructures, tourism complexes, and the interruption of activities caused by the closure of roads and the railway. One of the main strengths of SD modeling is that the model's flexibility allows for the inclusion of all causes and consequences.

The developed model structure was then analyzed for feedback relationships between the causes and consequences. Figure 7 provides a simple example representing two different policies suggested by two different stakeholders. The figure also describes the basic steps used by these stakeholders to develop their CLDs. In the first feedback loop (riverbed maintenance), the Bradano flood is marked as the problem variable. Maintenance of the riverbed and presence of aggregates and gravel are marked as direct and indirect causes, while damage to productive activities and city planning modification were added as direct and indirect consequences. Finally, city planning modification was linked with the presence of aggregates and gravel through a feedback loop, which connects the consequence with the cause and indicates the dynamics of the system. "*Policy A*," control of riverbed maintenance, is represented by a balancing loop which indicates that this policy if implemented, will balance out the current impacts of the presence of aggregates and gravel in the future (see Fig. 7).

In the second feedback loop (protection plan), flood events increase economic losses, which in turn cause a reduction in recreational activities. Reduction in recreational activities forces the government to adopt some protection plans. These plans cause a reduction in floodplain activities and thereby reduce flooding. "*Policy B*" in Fig. 7, named "protection plan," will have a reinforcing effect, meaning that this policy, if implemented, would contribute to a reduction in Bradano River flooding.

Each stakeholder prepared their CLD independently. During the entire process, the facilitator helped solely with procedural issues and remained neutral to ensure the development of unbiased qualitative mental models. Once stakeholder views in the form of causes, consequences, and feedback loops were expressed in the CLD, stakeholders were asked by the facilitator, “What kind of policies do you think would be effective in the mitigation of the flood problem in the study area?”. All stakeholders agreed that better management of the riverbed could help reduce the amount of flood damage, which they further indicated as a cost-effective management solution. Almost all stakeholders agreed that proper cleaning of the drainage channels was necessary to avoid the accumulation of vegetation, which often prevents the normal flow of water, causing an increase in flood intensity. These kinds of management options were identified as short-term policies.

Some stakeholders believed that the role of the dam was insignificant in past flooding events, while others pointed to poor dam management and suggested that past events were triggered by the opening of the spillway. As this situation required further analysis through quantitative modeling exercises, both options were included in the merged model diagram.

Some stakeholders believed that damage could be avoided by changing the type of crop cultivated in the riverbank areas. Furthermore, depending on the type of crops cultivated, farmers could benefit from this policy since the water could also help in the production or maturation of some products. This was considered to be a long-term policy.

Some local authorities proposed the construction of a levee on the left side of the river, with a storage area to collect drainage water from any overflow of the territory’s drainage channels, given their poor maintenance.

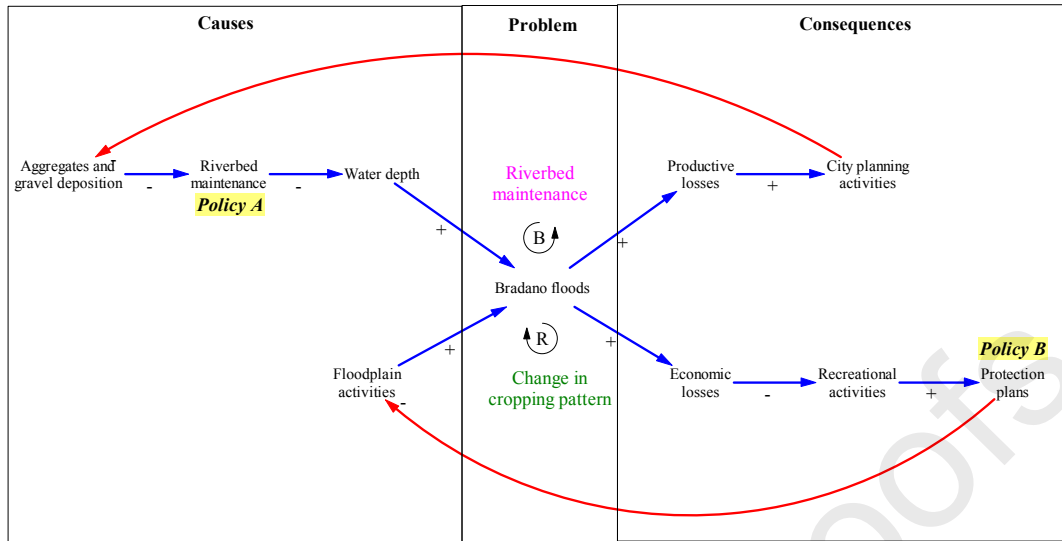


Fig. 7 – Example of balancing loops, reinforcing loops, feedback loops and policies in the construction of the CLD. Note: the letter B stands for balancing loop and R for reinforcing loop.

4.4 CLD Merging

In the current research, individual stakeholder CLDs were merged. The final combined CLD represents a holistic view of the complete system. The merging process was initiated with the most comprehensive model (*i.e.*, the model with the maximum numbers of variables). The process continued with the incorporation of additional variables until each of the stakeholder viewpoints were represented. All controversies and conflicting ideas were indicated in the merged diagram for future discussions when all stakeholders were present. Stakeholders were asked to attend one follow-up group meeting to finalize the merged CLD. Wide-ranging discussions regarding the group's opinions on the details of the merged CLD and any controversies were encouraged during this meeting.

This process also allowed stakeholders to share their ideas. The preliminary merged diagram (*i.e.*, the overall CLD) (Fig. 8) generated by moderators and, then, simplified as described in

section 4.6 was in a later stage presented to stakeholders to elicit their opinions regarding the overall model.

4.5 Merged CLD sub-processes and management options

The complete merged model consists of seven loops (Fig. 8) covering details and qualitative dynamics of different sub-processes such as agricultural activity, economic damage, dam maintenance, and infrastructure maintenance. Several processes, policies, and both local and global feedback loops can be identified.

Loop 1 (the 'Floodplain activity loop', indicated in green in Fig. 8) shows the balancing effect of the Bradano floods on residential structures close to the river. An increase in flooding leads to an increase in residential damage, which, in turn, leads to a reduction in structural value, thereby reducing floodplain activities.

The presence of activities in flooded areas makes the soil impermeable and, in case of an event, leads to a greater expansion of the floodplain. Conversely, flooding implies a greater water residence time within the watershed and possible water contamination, leading to a reduction in agricultural activities within the study area. This, in turn, leads to better maintenance of the levees by reducing the constant presence of farmers on adjacent land, which inevitably has a negative impact on their maintenance (loop 2 – 'Maintenance of the levees loop' indicated in orange, Fig. 8). Agricultural activities also have a negative effect on maintenance of the drainage channels located close to the river, as farmers keep these areas clean for cultivation, but, in doing so, move large amounts of vegetation into the channels, causing a narrowing of the river section, which reduces the outflow capacity of the riverbed (loop 3 – 'Maintenance of the drainage channels loop' indicated in orange, Fig. 8).

In terms of economic damage, stakeholders highlighted the need to review flood compensation and insurance policies and to apply new laws to avoid the reoccurrence of

disagreements that took place after past flood events. Stakeholders also identified that new laws should allow the removal of environmental constraints from business and public properties, so that authorities can carry out routine maintenance of the riverbed, thereby reducing the Manning's roughness coefficient and, consequently, the water depth in the river (loop 4 – 'Economic damage loop' indicated in black, Fig. 8).

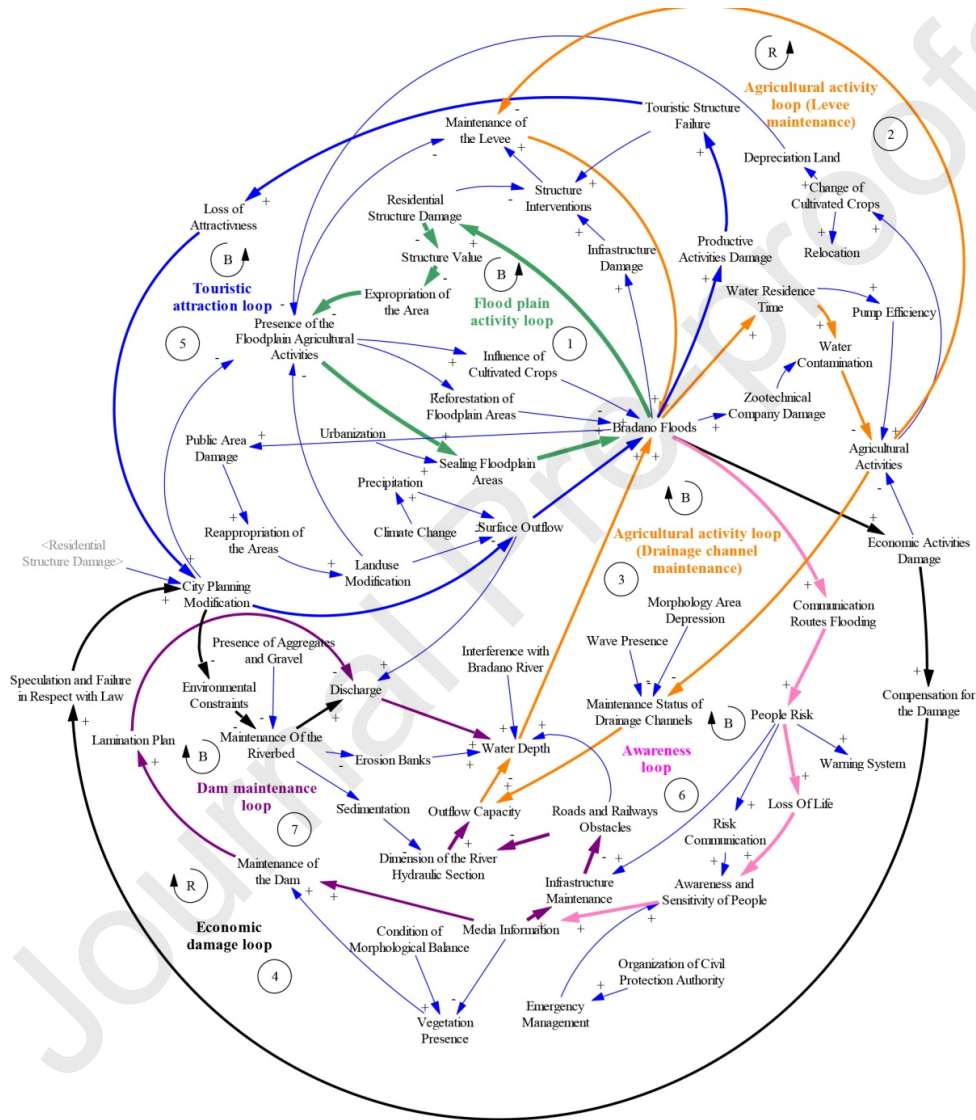


Fig. 8 – Final merged CLD developed from individual stakeholder CLDs

Loop 5 exhibited a balancing effect, as the Bradano floods initially cause a decrease in the attractiveness of riverside tourism, which leads to changes in urban planning to facilitate the

protection of cultural heritage and tourism facilities (loop 5 – 'Tourism attraction loop' indicated in blue, Fig. 8). The balancing nature of the loop indicates that tourism will decrease for only a short duration, and that after a change in urban planning to protect the region's cultural heritage, the tourism in the study area will recover.

Some stakeholders thought that greater awareness could lead to better information about hydraulic risk. Greater awareness of flood risk can be leveraged to pressure government agencies to improve the maintenance of infrastructure and to avoid constructing roads and railways that act as obstacles to river flow. These improvements would lead to an increase in the dimension of the hydraulic section, which in turn would cause a reduction in water depth (loop 6 – 'Awareness loop' indicated in magenta, Fig. 8).

Information from the media has a positive impact on dam management, which leads to a reduction of the discharge in case of a flood-wave, and contributing to reduced water depth. The dam maintenance loop addresses the reservations of some stakeholders who were convinced that the dam was the main cause of the 2011 flood event. They believed that heavy rainfall led to the opening of the dam's bottom spillway, flooding a large area and causing damage to agricultural activities, residential structures, tourism complexes, and infrastructures (loop 7 - 'Management of the dam loop' indicated in fuchsia, Fig.8).

4.6 Simplification of the merged diagram

The merged final diagram, presented in Fig. 8, was complex and difficult for stakeholders to understand. Stakeholders failed to identify important feedback loops and faced difficulty in capturing the underlying themes of the fundamental version of the working model. Therefore, the complex merged diagram was simplified using the CLD simplification approach recommended by Bureš (2017). The main purpose of this simplification was to facilitate stakeholder understanding of the system. However, the authors acknowledge that the

simplified CLD might fail to represent correct system boundaries and behavior due to the elimination of some important processes and variables. Hence, in a companion study, the final merged model (Fig. 8) will be used as a base model for developing a quantitative SD simulation model.

Fig. 9 shows a simplified graphic representation of the diagram. A dashed arrow is associated with a positive link while a normal fill arrow represents a negative link. The connection between dam maintenance and water depth is marked with an exclamation point, which means that the stakeholders have conflicting opinions regarding the effects of dam maintenance on water depth. Important activities were identified as floodplain activities and agricultural activities. These activities negatively affect levees and drainage channel maintenance within the basin, resulting an increased flood risk. Other important loops are the awareness and dam maintenance loops. As the dominance of awareness loop increases, dam maintenance increases, and vice versa. The simple CLD also highlights effects of flooding on tourism and economic growth in the area.

The simplification process was carried out through three main activities, namely endogenization, encapsulation, and order-oriented reduction as suggested by Bures, 2017.

Finally, the simplified diagram (Fig. 9) was introduced to all eight stakeholders who found that it was an appropriate and acceptable representation of the system. It is evident that the reduced diagram (Fig. 9) is simpler than the pre-reduction CLD (Fig. 8). The number of loops and the main themes are the same in both figures, but it was possible for all the stakeholders (in particular, non-experts) to understand the simplified diagram (Fig.9), and to focus on the real issues addressed in the case study. Furthermore, the reduced number of variables helped to avoid misunderstandings and misinterpretations. In the reduced CLD, water depth remains the primary causative factor in flooding. However, other significant variables are highlighted,

such as the maintenance of the riverbed, the presence of floodplain agricultural activities, and city planning modifications.

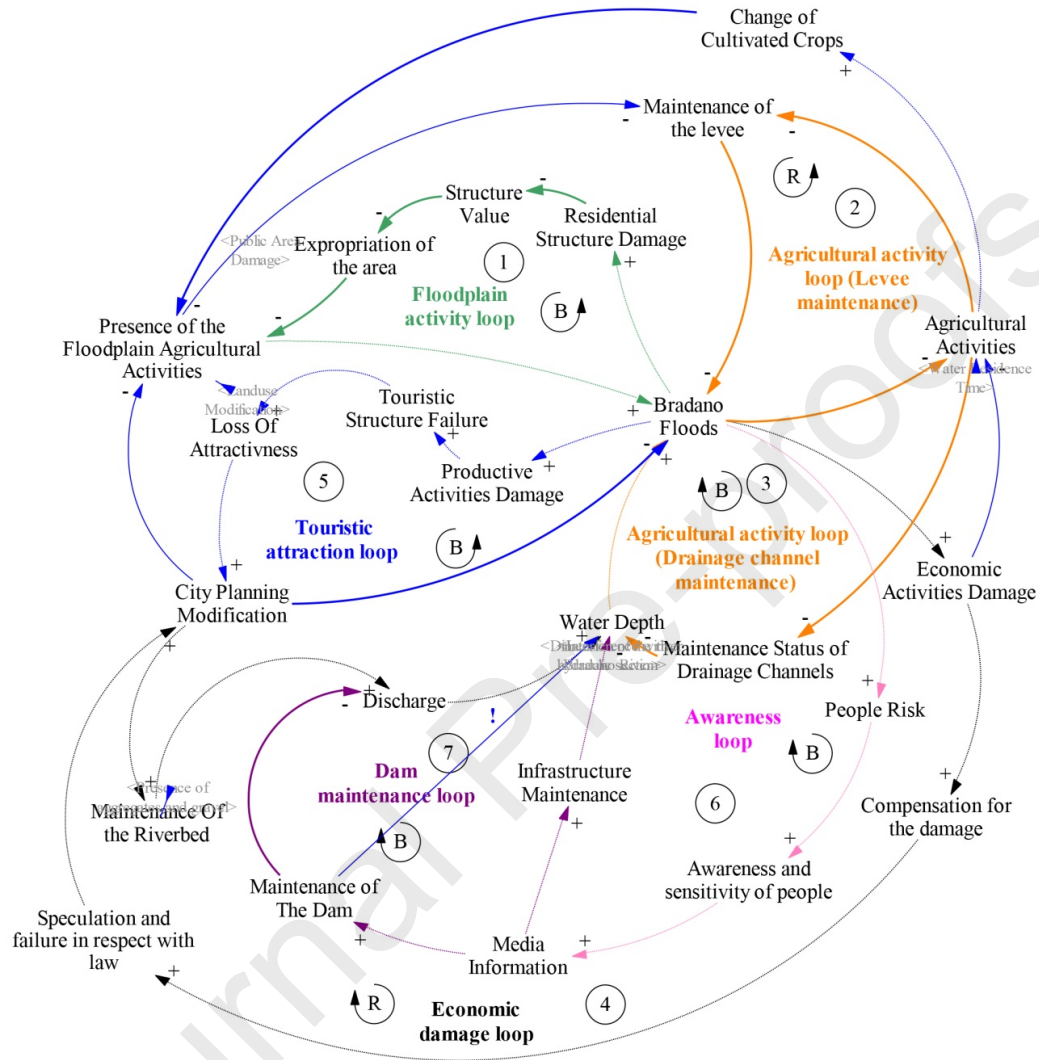


Fig. 9 – Reduced merged CLD

5. Discussion

The purpose of the current study was to demonstrate the usefulness of applying CLDs to involve key stakeholders in the initial stages of socio-economic-environmental-model development, facilitate stakeholder discussion, and develop an integrated perspective on complex issues in flood risk management. The methodology proposed in the study can help

bridge the communication gap between policymakers and local stakeholders. Other benefits include, but are not limited to, increasing social awareness about a problem of interest, improving understanding of socio-economic and environmental interactions, building knowledge of a complex holistic system using a simple approach, and improving the role of stakeholders in decision-making.

Regarding the specific case study analyzed, stakeholders highlighted the importance of the additional inputs provided by the present study to management policies for flood risk mitigation in the study area. Recent and planned mitigation actions (*i.e.*, the San Giuliano dam lamination plan and an extension of the railway bridge-tunnel, respectively) were implemented as a result of media pressure or based only on technical judgment with limited involvement of local stakeholders. This situation highlights the importance of participatory modeling to incorporate the knowledge of key stakeholders for a holistic view of the complex system. This study is first of its kind for flood risk management in the Bradano River.

Overall, stakeholders appreciated the flexibility of the CLD method and found it to be useful, cost effective, and simple to understand. They further agreed that final merged diagram represented a holistic view of dynamic system processes, and therefore can be used to increase awareness and understanding of the interactions between different system components. Furthermore, they appreciated the transparency of the process. The approach was easily understood by non-modeling professionals and provided them with support during the dialogue processes. Stakeholders were able to consider different management policies, and the results revealed socio-economic and territorial aspects of the issue that had not been considered by other studies. In particular, farms believe the final CLD will help to increase the collaboration between them and the decision-makers turning decision-makers' attention to their opinions. The farmers state, also, that final CLD has increased their comprehension of the interactions between causes and consequences of the systems. Moreover, the Water Basin

Authority believes that the proposed approach could support them in their compliance with the EU Flood Directive that require that require the public participation process in the preparation of flood management plan (e.g. 2007/60/EC Art.10). Few others were skeptical, from the beginning, that CLD will effectively affect the future decisions of decision-makers but they remained impressed about the clear framework showed by the CLD considering the minimum efforts from their side. This demonstrates the usefulness of the proposed approach, which successfully fostered policy coordination between stakeholders from different sectors, groups, levels, and interests. The model was developed with broad boundaries and included diverse socio-economic and physical variables. For example, components that were included in the current model such as land-use changes, damage compensation, floodplain activities, and stakeholder awareness, had not been previously investigated in this region.

The present study addresses the need for a modeling tool that facilitates understanding of the dynamics of flood risk management from both environmental and socio-economic, perspectives. Indeed, this paper covers the first phase of a long-term study in which the authors are involved. The scoping phase, *i.e.*, qualitative modeling using CLDs, is described in the current paper, while in subsequent papers, model quantification, testing, and institutionalization will be discussed. In particular, the present paper examines the development of a qualitative SD modeling approach for stakeholder engagement in flood risk management.

In a planned study, the group-built CLD (Fig. 8) discussed in this manuscript will be used as a base model for developing a quantitative SD simulation model to simulate the socio-economic aspects and technical measures of flood risk management in an integrated way. Finally, the linking (dynamic coupling) of more detailed, physically-based models and group-built SD models will be an important future research focus for flood risk management in the Bradano River, along with other locations. The results of the case study indicate that CLDs

are an effective and simple method to initialize stakeholder engagement. The method was helpful in the development of qualitative models aimed at addressing complex issues, such as flood risk management. Qualitative models enable stakeholders to share their opinions and learn from each other to develop a more in-depth and holistic understanding of the system under investigation.

6. Conclusion

This study is the first of its kind aimed at an integrated analysis of the issue of flood risk management with the active participation of stakeholders through CLD modeling. The results of the case study indicate that the proposed CLD method was helpful in the development of qualitative mental models aimed at addressing complex issues, such as flood risk management. Stakeholders were easily engaged due to the simplicity and limited costs (in terms of time) of the interview process. They each created individual CLDs without any previous experience with SD modeling and, in a few cases, with limited technical skills related to the topic. The individual interview format created an open environment for stakeholders to share their opinions.

Moreover, all stakeholders were able to understand the simplified version of the merged group CLD. The participating stakeholders were satisfied with the modeling process and the simplified group CLD model. Many were surprised by the transparency of the process and enthusiastic about how well they were able to depict aspects of a complex system in a clear, simple, and understandable way. In particular, the overall group CLD model highlighted problems in the study area that had not been considered in previous studies. The CLD exercise allowed key stakeholders from different organizations and groups to learn from each other and develop a more in-depth and holistic understanding of the system under investigation, leading to increased social awareness, group learning, and consensus building.

The knowledge of the problem obtained through the participatory modeling approach went beyond the information that was gathered from the literature review and pre-existing hydraulic studies.

The proposed methodological framework was applied to initialize stakeholder engagement in flood risk management in the Bradano River Delta in Italy. However, the modular structure and the universal nature of the employed methods make the proposed scheme scalable, supporting its application in other territorial contexts. The above-mentioned characteristics of the proposed approach also permit its extension to rainfall-induced shallow landslides, *i.e.*, the most common natural hazards in some areas of the world (Peruccaci et al., 2017; Manenti et al., 2019), through a correct and clear definition of the problem, key variables, and time horizon. Water-related risks are complex and dynamic issues. The methodology proposed here can be used to construct a holistic representation of the risk system by analyzing the behavior and the connection between different socio-economic variables and integrating stakeholder perspectives.

In future research, the CLDs constructed by stakeholders will be quantified in a SD modeling environment and coupled with a physically-based model to evaluate the consequences of the stakeholders' proposed policies. By considering stakeholder perspectives, as well as social, environmental, economic, and physical aspects of the problem, effective flood risk management decision-making in the region is supported.

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- A cost-efficient methodology to initialize stakeholder engagement in flood risk management was proposed.
- The methodology was tested in a case study in the Bradano River Delta in Italy
- The methodology is based on causal loop diagrams using a system dynamics modeling approach.
- The methodology introduces alternative ethical perspectives to encourage more balanced decision-making.
- The results revealed important, and not yet studied, causes of flood risk and solution strategies.

Participatory methods to support successful policy decisions regarding the complex and dynamic interactions of social, ecological, and physical processes involved in flooding must be simple, easy-to-use, and cost-efficient. Accordingly, a stepwise methodology framework based on causal loop diagrams (CLDs) was developed to address the challenge of context-sensitive initialization of key stakeholders in the collaborative flood risk management process. The methodology consists of five main stages: (i) problem definition, (ii) stakeholder analysis and identification of key groups, (iii)

interviews with key stakeholders to construct individual CLDs, (iv) merging of individual CLDs to form a holistic qualitative model representing the entire system, and (v) implementation of an order-oriented reduction process to simplify the final merged CLD, thereby increasing understanding of the most important processes and feedbacks. The proposed approach for flood risk management was tested in a coastal area of southern Italy, a region historically affected by flood events. Given its simplicity, the proposed method emerges as a valuable tool to elicit and map mental models, especially when working with the most inexperienced stakeholders, *i.e.*, farmers, agricultural companies, tourist complexes holders, and infrastructure authorities. Indeed, all stakeholders in the current study were able to understand the process and proposed different flood risk management policies, such as land-use changes, management of damage insurance payouts, floodplain activities, and improvement of public awareness. The proposed methodology overcame multiple barriers in initializing stakeholder engagement, including the technical focus of most flood management agencies, the additional cost and time requirements for stakeholder involvement, as well as institutional structures that impede collaborative management. Moreover, the results point to socio-economic aspects of flood risk management that have not been considered in previous modeling studies.